WORKED SOLUTION

STEP II 1998, Question 4

Pure • Integration by Parts

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q4

The key idea

The awkward factor $\operatorname{cosec}(x/2)$ can't be integrated directly, but the *difference* of two consecutive integrands is much kinder: the sine difference formula produces a factor $\sin(x/2)$ that cancels the cosecant exactly. What is left is a polynomial times a cosine, which is a standard integration by parts. Then the differences telescope down to a starting value we can compute by hand.

Solution

The difference $I_n - I_{n-1}$

Since $nx + x/2 = (n + \frac{1}{2})x$, the two integrands differ by

$$\left(\frac{\pi}{2} - x\right)[\sin(n + \tfrac{1}{2})x - \sin(n - \tfrac{1}{2})x] \operatorname{cosec} \tfrac{x}{2}.$$

By the factor formula $\sin A - \sin B = 2\cos \frac{A+B}{2} \sin \frac{A-B}{2}$, with $\frac{A+B}{2} = nx$ and $\frac{A-B}{2} = \frac{x}{2}$,

$$\sin(n + \tfrac{1}{2})x - \sin(n - \tfrac{1}{2})x = 2\cos nx \sin \tfrac{x}{2}.$$

The $\sin \frac{x}{2}$ cancels the cosecant, so

$$I_n - I_{n-1} = 2\int_0^{\pi} \left(\frac{\pi}{2} - x\right) \cos nx \, \mathrm{d}x.$$

Ask yourself Is the cancellation legitimate at $x = 0$, where $\operatorname{cosec}(x/2)$ is infinite?

Yes. Each original integrand has a finite limit as $x \to 0$, because $\sin(n + \frac{1}{2})x / \sin \frac{x}{2} \to 2n + 1$, so the integrals are perfectly ordinary. For $0 < x \leq \pi$ the cancellation is exact algebra, and changing an integrand at the single point $x = 0$ does not change the integral.

Now integrate by parts with $u = \frac{\pi}{2} - x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = \cos nx$, so $v = \frac{1}{n}\sin nx$ (valid as $n \geq 1$):

$$\int_0^{\pi} \left(\tfrac{\pi}{2} - x\right)\cos nx \,\mathrm{d}x = \left[\left(\tfrac{\pi}{2} - x\right)\frac{\sin nx}{n}\right]_0^{\pi} + \frac{1}{n}\int_0^{\pi} \sin nx \,\mathrm{d}x$$
$$= 0 + \frac{1}{n}\left[-\frac{\cos nx}{n}\right]_0^{\pi} = \frac{1-\cos n\pi}{n^2} = \frac{1-(-1)^n}{n^2},$$

where the boundary term vanishes because $\sin 0 = \sin n\pi = 0$. Therefore

$$I_n - I_{n-1} = \frac{2(1-(-1)^n)}{n^2} = \begin{cases} \dfrac{4}{n^2}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

Evaluating I_n

To telescope we need a starting value. The integral defines I_0 just as well: with $n = 0$ the integrand is $(\frac{\pi}{2} - x)\sin\frac{x}{2}\operatorname{cosec}\frac{x}{2} = \frac{\pi}{2} - x$, so

$$I_0 = \int_0^{\pi}\left(\tfrac{\pi}{2} - x\right)\mathrm{d}x = \left[\tfrac{\pi}{2}x - \tfrac{1}{2}x^2\right]_0^{\pi} = 0,$$

and the difference formula above holds for $n = 1$ too (the working used only $n \geq 1$). Adding the differences for $1, 2, \ldots, n$:

$$I_n = I_0 + \sum_{k=1}^{n}(I_k - I_{k-1}) = \sum_{k=1}^{n}\frac{2(1-(-1)^k)}{k^2}.$$

Only odd k contribute, each giving $4/k^2$, so

$$\boxed{I_n = 4\sum_{\substack{k=1 \\ k \text{ odd}}}^{n}\frac{1}{k^2} = 4\left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots\right),}$$

the last term being $1/n^2$ if n is odd and $1/(n-1)^2$ if n is even. Equivalently, $I_n = 4\sum_{r=1}^{m}\frac{1}{(2r-1)^2}$ with $m = \lceil n/2 \rceil$.

If you are uneasy about using $n = 0$ (the question says n is positive), start from I_1 instead. Since $\sin\frac{3x}{2} = \sin\frac{x}{2}(1 + 2\cos x)$, we get $I_1 = \int_0^{\pi}(\frac{\pi}{2} - x)(1 + 2\cos x)\,\mathrm{d}x = 0 + 2 \cdot 2 = 4$, using the integration by parts above with $n = 1$. This agrees with the formula.

Checks

- $I_1 = I_2 = 4$ and $I_3 = I_4 = 4 + \frac{4}{9} = \frac{40}{9} \approx 4.4444$. Simpson's rule on the original integral (with the integrand's limit $\frac{\pi}{2}(2n+1)$ at $x = 0$) gives $4.444444\ldots$ for I_3, and agrees with the sum to 10^{-8} for every $n \leq 12$.
- The integrand for n is $\frac{\pi}{2} - x$ times $1 + 2\cos x + \cdots + 2\cos nx$, and $\frac{\pi}{2} - x$ is odd about $x = \frac{\pi}{2}$. For even k (and for the constant 1), $\cos kx$ is even about $x = \frac{\pi}{2}$, so the product is odd and integrates to zero over $[0, \pi]$. That is why $I_0 = 0$ and even k contribute nothing.

Common slips

- Getting the sign or the factor 2 wrong in the sine difference formula; test it at a value such as $x = \pi$, $n = 1$.
- Writing $\cos n\pi = 1$ instead of $(-1)^n$, which makes every difference zero.
- Forgetting the starting value, or writing the answer as a sum over all k without noticing that the even terms vanish.

Beyond STEP

The function $\sin(n + \frac{1}{2})x \big/ \sin\frac{x}{2} = 1 + 2\sum_{k=1}^{n} \cos kx$ is the *Dirichlet kernel*, the central object in proving that Fourier series converge. As $n \to \infty$ our sum tends to $4 \cdot \frac{\pi^2}{8} = \frac{\pi^2}{2}$, using $\sum_{\text{odd } k} 1/k^2 = \pi^2/8$, a close relative of Euler's $\sum 1/k^2 = \pi^2/6$ (numerically, $I_{200} \approx 4.925$ against $\pi^2/2 \approx 4.935$). Fourier series and these sums are covered well in Boyce and DiPrima, *Elementary Differential Equations*, and the analysis behind the convergence in Abbott, *Understanding Analysis*.

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